

Easwaran Subramanian

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INTRODUCTION

Data and ML engineer with 3+ years of experience building data pipelines, forecasting models and production ML systems. Skilled in Python, SQL, PySpark and cloud-based deployment, with a focus on reproducible workflows and automated monitoring. Highly fluent with modern AI coding tools and agentic workflows, using them regularly to build things without compromising code quality.

Recently completed a Master's in Machine Learning at the University of Tübingen.

SKILLS

Programming: Python, SQL, Bash

Data Engineering: ETL, PySpark, Pandas, NumPy

Databases: PostgreSQL, MySQL, Relational Data Modelling

Cloud & Infrastructure: GCP, AWS - EC2/S3/Lambda/Step Functions/Athena/Glue, Docker, Linux

Machine Learning: PyTorch, Scikit-learn, Hugging Face

Monitoring & CI/CD: GitHub Actions

AI Tools: Claude Code, Codex

Languages: English (C1), German (A2.2), Tamil (Native)

WORK EXPERIENCE

Graduate Data Researcher | Tübingen Institute for Medical Education, Tübingen

NOV 2023 – JUL 2024

- Built a Python ETL pipeline to ingest, clean, validate, and aggregate survey data from 200+ medical professionals, replacing a multi-hour manual process with a one-click workflow.
- Built a PHP-based internal tool that let researchers configure processing parameters and regenerate analyses without developer support.
- Redesigned MySQL tables and optimized queries, improving data retrieval speed while preserving controlled access to sensitive records.

Machine Learning Engineer | Sony B.V., Stuttgart (Internship)

MAR 2023 – AUG 2023

- Built reusable Python and PyTorch workflows for data preprocessing, model training, evaluation and hyperparameter search.
- Contributed to an automated Neural Architecture Search pipeline that evaluated 20+ model configurations, helping identify one that improved performance by ~10%.
- Wrote test classes, standardized preprocessing, configuration and evaluation steps to improve experiment reproducibility.

Data Analyst | Verloop.io, Bengaluru, India

MAR 2022 – SEP 2022

- Built Python data pipelines to process, normalize and cluster thousands of chatbot conversations for downstream analytics and model-improvement workflows.
- Automated the extraction, transformation and aggregation of conversation data for a live reporting dashboard, cutting the KPI reporting cycle by ~30%.
- Defined reusable processing logic for mapping unstructured user expressions to standardized intent categories, improving data consistency and increasing intent-classification F1-score from 0.78 to 0.85.

- Built Python and SQL workflows to prepare market, pricing and demand data across 15+ markets for forecasting and optimization.
- Built a demand-forecasting and pricing model using conjoint analysis, contributing to a reported 10-15% revenue improvement.
- Implemented constrained optimization (COBYLA) to generate pricing recommendations under business rules, including price ceilings and minimum sales volumes.
- Automated evaluation of 100+ pricing scenarios, measuring expected effects on revenue, margin and demand for business stakeholders.

RECENT PROJECTS

Edge-SLM: Training and Serving a Small Language Model

- Building a versioned DVC pipeline covering synthetic-data generation, preprocessing, training and model artifact management, with the model deployed via FastAPI and monitored using Prometheus for latency and inference performance.
- Setting up GitHub Actions to run automated benchmarks and block changes that fall below performance thresholds.
- Currently quantizing a 0.5B-parameter model to 4-bit, targeting a reduction in memory footprint from 1.1 GB to ~280 MB.

Quantifying Neural Adaptability: A Reproducibility Study of Hypencoder [MASTER'S THESIS]

- Replicated the Hypencoder architecture (a neural information retrieval model), using hypernetworks to generate query-specific ranking weights for document scoring.
- Tested scaling limits by distilling knowledge from stronger cross-encoders and BM25 to identify scaling bottlenecks in pooled representations.
- Ran SLURM-orchestrated training on the MS MARCO benchmark to reach 0.727 nDCG@10 and improved zero-shot accuracy.
- Code: https://github.com/Eashwar-22/Neural_Adaptability_Reproduction_of_Hypencoder
- Paper: <https://drive.google.com/file/d/1sBa5Jxoj1qcTB2kal3MMl2v9GYkXqBzA/view?usp=sharing>

Adaptive LLM Router

- Built a FastAPI gateway that routes traffic between local (student) models and cloud (teacher) models, cutting external LLM API usage by 70% by offloading intent-classified tasks locally - a pattern similar to production systems like RouteLLM.
- Continuing to optimize a ModernBERT classifier and semantic cache for on-device inference using ONNX/CoreML.
- Currently adding PII redaction (Microsoft Presidio) and automated response-quality evaluation using DeepEval (LLM-as-a-judge), with a dashboard for tracking results.
- Code: https://github.com/Eashwar-22/adaptive_llm_router

EDUCATION

Msc. in Machine Learning | University of Tübingen, Baden-Württemberg, Germany

OCT 2022 – MAR 2026

B.Tech in Electronics and Communication Engineering | Vellore Institute of Technology, TN, India

JUN 2016 – JUL 2020